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## CUBOID CONSTRUCTION ELEMENT, METHOD FOR MANUFACTURING A CONSTRUCTION ELEMENT, AND SYSTEM COMPRISING TWO CUBOID CONSTRUCTION ELEMENTS

### Abstract

The invention relates to a substantially cuboid construction element (**101, 102**) made of plastic for use as a construction material. The construction element (**101, 102**) comprises four side surfaces (**21, 22, 23, 24**) which are vertically oriented when the construction element (**101, 102**) is in an installation position (**11**). The side surfaces (**21, 22, 23, 24**) each have at least one receiving structure (**3**). The upper side (**4**) of the construction element (**101, 102**), in the installation position (**11**), has at least one male connecting element or one female connecting element, and a lower side (**5**) of the construction element has at least the other male connecting element (**6**) or female connecting element (**7**). The male connecting element (**6**) and female connecting element (**7**) are shaped so as to be complementary to one another.

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## Background/Summary

[0001] The invention relates to a box-shaped component, a method for manufacturing a substantially box-shaped component, and a system comprising at least two box-shaped components and a connecting element according to the independent claims.
[0002] The construction of buildings often requires specialist knowledge, meaning that the cost of services often exceeds the cost of materials many times over. Private individuals can generally only play a minor role in the construction of buildings, as the arrangement and alignment of building materials, especially load-bearing structures such as walls and foundations, is only possible with a great deal of labor and incorrect arrangements are difficult to correct. In addition, construction projects using conventional building materials often require a wide range of equipment to which private individuals do not have access, or which at least entail substantial financial costs.
[0003] Repair work and/or extension projects are often associated with considerable expense and/or cannot be carried out flexibly according to your own design ideas.
[0004] In addition, the cost of building materials is often very high, preventing affordable housing from being accessible to a large number of people.
[0005] Various elements for use as building materials are known in the state of the art. However, there is generally a lack of a simple, intuitive, user-friendly and quick method of use for arranging the components relative to each other and fixing the components, in particular without additional equipment.
[0006] For example, US 2004/000114 A1 shows modular components that have passages so that the components can be fixed relative to each other. However, the use of these components is complicated and requires concrete and structural steel for fixing.
[0007] It is the task of the present invention to overcome these and other disadvantages of the prior art. In particular, a simple and cost-effective component for use as a building material is to be provided.
[0008] This task is solved by a component defined in the independent claims, a method for manufacturing a component, and a system. Further embodiments result from the dependent claims.
[0009] A substantially box-shaped component according to the invention, comprising at least partially plastic and four side faces oriented vertically in an installation position. The side faces each have at least one receiving structure. In the installation position, an upper side of the component has a male connecting element and a lower side has a female connecting element or the upper side has a female connecting element and the lower side has a male connecting element.
[0010] The male connecting element and female connecting element are shaped to complement each other.
[0011] Shaped complementary to each other defines in this context an at least partially form-fitting connection possibility of the elements shaped complementary to each other.
[0012] The component can consist of or include thermosets or thermoplastics.
[0013] In particular, the component may consist of or comprise thermosetting plastics such as cured synthetic resins cured prepolymers.
[0014] In particular, the component may consist of or comprise thermoplastics such as polyolefins, in particular preferably polypropylene and/or polyethylene, polymethyl methacrylate, or polystyrene.
[0015] The component can also consist of 50%, preferably 90%, in particular preferably essentially

completely, of plastic.
[0016] The component can also include fillers and/or colorants. Inexpensive fillers, such as quartz sand, can reduce the cost of the components. Dyes can be adapted to the aesthetic preferences of the user or used to mark different components in order to improve recognizability.
[0017] The component can be made from at least partially recycled plastic to reduce costs.
[0018] The component can include fire protection, thermal insulation and/or noise insulation materials. The trapped air inside the component also has an insulating effect. In addition, insulating material can be attached outside or inside the component. This has the advantage that the additional step of attaching insulation can be avoided.
[0019] Alternatively, separate insulating materials can be attached to the component, preferably through a casing or between a casing of the component.
[0020] The component can be used for the construction of walls of buildings.
[0021] A component can have a horizontal recess in the installation position to accommodate a support device, in particular beams. The horizontal recess can run through the component completely or only partially. This means that components on two opposite sides of a wall can accommodate a support device that can be used, for example, to support a false floor or a ceiling, in particular false floor elements.
[0022] The at least one male connecting element and the female connecting element can be aligned in the longitudinal direction perpendicular to the lower side and/or upper side. In particular, the at least one male connecting element can be arranged on the upper side in an installation position of the component. Due to such alignment and arrangement, the components can be easily stacked without slipping and can also be transported in a space-saving manner.
[0023] The male and female connecting element can have a round, oval, polygonal, in particular regular polygonal shape in cross-section.
[0024] An oval, polygonal or regular polygonal shape of the connecting elements has the advantage that two different components cannot be rotated when the male and female connecting elements of two components are connected. This makes it easier to arrange the components in a straight line one above the other.
[0025] The upper side and/or lower side can each have one to ten connecting elements, in particular one to ten male or one to ten female connecting elements.
[0026] The component can be mirror-symmetrical along one or two planes. Such mirror symmetry simplifies the arrangement of the components in relation to each other for the user.
[0027] The receiving structure can extend continuously in a straight line from an upper side of the component to a lower side of the component. Preferably, the receiving structure extends essentially in a vertical direction in the installation position of the component.
[0028] The receiving structure may have at least one recess extending over part or all of the side face. The cross-section of the recess can include an undercut.
[0029] The cross-section of the recess can be L-shaped, T-shaped, C-shaped, Z-shaped, dovetail-shaped, trapezoidal, knob-shaped, oval and/or curved. Such a cross-section enables two components to be connected, preferably by essentially positive connections, by inserting a connecting element, in particular a connecting bar. Such a connection can counteract a force acting perpendicular to the connected side faces, which pushes the two components apart.
[0030] The receiving structure can have at least two recesses, in particular three, four, five or six recesses. The longitudinal axes of the recesses can be parallel and spatially spaced apart.
[0031] The undercuts of the recesses can run in cross-section in the direction of the adjacent recess. Such undercuts allow the recesses to be spaced further apart on the side face. The distance between two recesses can be between 2 cm and 30 cm, in particular 20 cm. A greater distance between the recesses means that the transverse forces acting on components or connecting elements can be better compensated.
[0032] The receiving structures can have a recess which has a straight longitudinal axis and which

runs partially or completely parallel to the recesses in the longitudinal direction. This recess can be suitable for receiving a connecting bar of an attachment element, so that the connection between two components can be additionally reinforced. The receiving structure can have a recess in addition to or as an alternative to the recess. The recess can be designed to accommodate a connecting bar and/or a cable.
[0033] A further task of the present invention is to provide a component which allows cables to be laid in the wall and at the same time provides an aesthetically pleasing, aligned wall surface.
[0034] The box-shaped component can be a component as described above. The component comprises four side faces aligned vertically in an installation position of the component.
[0035] At least two, preferably three or four, of the side faces each have a receiving structure.
[0036] At least two, preferably three or four, of the receiving structures comprise at least one recess for receiving at least one attachment element for connection to an adjacent component or a wall panel.
[0037] At least one of the receiving structures, preferably a receiving structure with a recess for receiving at least one attachment element, has a recess for receiving a cable. The recess is arranged, preferably centrally, in such a way that recesses of components stacked on top of one another are aligned, so that a continuous receptacle for a cable is formed.
[0038] The recess extends from one side face to an opposite side face. Such a recess can be particularly advantageous in combination with the embodiments explained above with receiving structures for receiving at least one attachment element on all four side faces. It is understood that such recesses for receiving a cable can also be advantageous in connection with other types of connection between individual components.
[0039] In the installation position, an upper side of the component can have a male connecting element and a lower side can have a female connecting element, or the upper side can have a female connecting element and the lower side can have a male connecting element.
[0040] The male connecting element and female connecting element can be shaped to complement each other.
[0041] The upper side of the component may comprise a step, preferably a step between at least one of the side faces and the upper side, so that components arranged one above the other form an at least partially enclosed cavity. The cavity is adjacent to a recess, preferably at least two recesses.
[0042] The step can extend at least partially or completely along a circumferential direction of the component. An arrangement of the step in the circumferential direction enables electrical cables to be routed in a horizontal direction in an installation position of the component.
[0043] The step can extend along an outer edge, formed by the upper side with the side faces of the component. This makes the step easily accessible and an attachment element, preferably with at least one connecting latch, can be easily attached.
[0044] The component may comprise a passage having a longitudinal axis which is aligned rectilinearly, preferably vertically, in the installation position and extends from the upper side to the lower side.
[0045] Such a passage offers the advantage that a water pipe, an electricity line, a radiator and/or a support structure can be placed in the passage.
[0046] A vertical passage in one installation position offers the advantage that several components arranged one above the other can form a continuous passage.
[0047] The cross-section of the passage can be circular, elliptical or polygonal, in particular regular polygonal.
[0048] The passage has a maximum diameter of 3 cm to 15 cm or essentially 30%, in particular 20%, of the maximum side length.
[0049] The passage can form the female connecting element.
[0050] The passage can extend through the protruding area of the male connecting element.
[0051] When connecting the female and male connecting elements of two components with one

passage each, an aligned connection of the passages to each other can be ensured.
[0052] The longitudinal axis of the passage can run through a mirror axis of the component, preferably through the geometric center of gravity of the component. This means that a user has to pay less attention to the orientation of the passage when connecting components.
[0053] The component can be mounted in the installation position [0054] a height, measured in vertical direction, in the range of 10 cm to 40 cm, in particular 15 cm to 25 cm; [0055] a width, measured from the farthest side faces, in the range from 10 cm to 80 cm, in particular 15 cm to 50 cm; [0056] have a depth, measured in a direction perpendicular to the height and width, in the range from 10 cm to 40 cm, in particular from 15 cm to 25 cm.
[0057] The height results from the vertical height of the component, excluding the height of the male connecting element.
[0058] The essentially box-shaped component can be produced by injection moulding.
[0059] A further aspect of the invention relates to a system comprising at least two components, in particular components as described above, and a connecting element for substantially positive connection to a receiving structure of one of the components and connection of two receiving structures of the components.
[0060] The other elements of the system may consist of or comprise the same materials as a component described above.
[0061] The attachment element can comprise a head section with a central web that has at least one lateral flange, preferably two lateral flanges.
[0062] The two lateral flanges can be arranged on opposite sides of the head section.
[0063] The head area of the attachment element can be at least partially rectangular, round, oval, dovetail-shaped, trapezoidal, double-T or I-shaped.
[0064] The head area of the attachment element in the installation position can have essentially twice the horizontal width of the step of the component, preferably along the entire longitudinal axis of the head area. Such a head area is advantageous, as the head area can thus be positively received by the steps of two components in order to fix two components relative to each other.
[0065] The center web can have a smaller width than the at least one lateral flange, in particular both lateral flanges.
[0066] The lateral flange can be materially connected to at least one connecting bar. A first lateral flange can be materially connected to a first connecting bar and a second flange can be materially connected to a second connecting bar. The at least one connecting bar can be inserted substantially positively into at least part of the receiving structures of two components at the same time in order to connect the two components.
[0067] At least two, preferably three, connecting bars can be materially connected by the head area so that they can be moved together, in particular can be inserted into the receiving structure by a translational movement. Thus, several recesses and/or recesses of a receiving structure of one component can be connected to another receiving structure of another component by a single attachment element.
[0068] The head area can be shorter in the horizontal direction than a horizontal side of the component. Such a head area has the advantage that power cables and heaters can still be attached horizontally in the step of the component from one side face to an opposite side face despite the attachment element being attached.
[0069] The connecting bolts can be fitted in the outermost area of the side flange.
[0070] The center bar can also have a connecting bar for insertion into the recess of the receiving structure of the component.
[0071] The connecting bars can run at least partially parallel to each other. This has the advantage that several connecting bars can be inserted into the recess(es) and/or recesses of two components by a single translational movement in order to connect them.
[0072] The cross-section of the connecting bar can have a complementary shape to two adjacent

recesses, in particular a double-L, double-T, double-C, double-Z, dovetail-shaped, trapezoidal, double-knob-shaped, double-oval or curved shape.
[0073] A dovetail-shaped connecting bar is more rigid and enables two components to be fixed in a stable position relative to each other with a single connecting bar.
[0074] Two connecting latches, in particular connecting latches of an L-shaped form, on the other hand, enable a more flexible connection of components. Even if a connecting latch is damaged or defective, such connecting latches can still be used to connect two components.
[0075] The steps, recesses and recesses of the components can essentially accommodate the attachment element with a positive fit. This simplifies the connection of two components with an attachment element and also minimizes the relative movement of the components to each other.
[0076] The system may comprise at least one first box-shaped component and at least one second box-shaped component. The first component may be substantially half the size, one third the size, or one quarter the size of the second component. The first component can be essentially cube-shaped.
[0077] This has the advantage that rows of components arranged on top of each other can be staggered and still be arranged laterally at corners, door frames and windows. A staggered arrangement of the components ensures greater stability.
[0078] The system may comprise a wall panel and the wall panel may have a first connection area on a first side face. The connection area can protrude from the side face and can be positively connectable to the receiving structure of the component.
[0079] The connection area can be shaped in such a way that the connection area can be connected to the recess of the receiving structure. However, the recess can remain free of blockages when the wall panel is connected to the receiving structure, so that electrical and/or water pipes can be routed through the recess while the wall panel is connected.
[0080] The wall panel can also protrude laterally over at least one side face of the component, so that two wall panels arranged at an angle to each other form a protruding corner.
[0081] A laterally protruding area of the wall panel can be mitered so that the corner joint can be arranged at an angle.
[0082] The first connection area of the wall panel can have a substantially complementary shape to the at least one recess and/or a depression, in particular L-shaped, T-shaped, C-shaped, Z-shaped, dovetail-shaped, knob-shaped, oval, and/or curved. This has the advantage that the receiving structure is suitable both for connecting adjacent components by means of attachment elements and for attaching wall panels.
[0083] The receiving structures, in particular the recesses and recesses, of components lying one above the other can be arranged in such a way that they run along a common longitudinal axis. Such an arrangement has the advantage that elements, such as a casing element, can be moved from an upper end of a receiving structure of a plurality of superimposed components to a lower end of the plurality of superimposed components along the receiving structure of the components.
[0084] The wall panels can be shaped in such a way that the connection area can only be attached to the at least one recess and at least one recess or recess of the component is not filled. This has the advantage that cables can be laid on a side face of the component and the wall panel can then be attached above it.
[0085] The connection area of the wall panel optionally does not extend over the entire extent of the component or has interruptions. This offers the advantage that cables, lines or wall heaters can be routed laterally to neighboring components in addition to the longitudinal axis of a recess. In particular, the cables, lines or wall heaters can be routed through a cavity created by the step of the component when stacking components on top of each other.
[0086] The wall panel can also have recesses to accommodate cables, pipes or wall heating.
[0087] The wall panel may have widened areas comprising noise or heat insulating material. Such wall panels have the advantage that insulation can be easily attached, replaced and, in particular,

easily retrofitted or upgraded.
[0088] The opposite second side face of the wall panel can form a substantially flat surface. The wall panel can then be used as a cover for the component.
[0089] The system may include a separate casing element. The wall panel may have a second connection area for attaching the casing element on a second side face opposite the first connection area.
[0090] The longitudinal axis of the second connection area can extend partially or completely in a horizontal direction in an installation position of the wall panel. In an installation position of the wall panel, the second connection area can be open at the top, in particular forming an at least partially hook-shaped cross-section. Such an arrangement has the advantage that casing elements can be fixed relative to the wall panel by gravity alone. In addition, the casing elements can thus be attached, as the casing elements can be inserted vertically from above into the second connection area.
[0091] The wall panel and/or casing element can have essentially the same periphery as a side face of the component.
[0092] However, a periphery of the wall panels and/or casing elements can also extend over the side face of several components.
[0093] The system can comprise a corner casing element. The corner casing element can have at least one connection area, so that the corner casing element can be connected to at least two receiving structures of adjacent side faces of a component. Alternatively, the connection area of the corner casing element can be connected to two second connection areas of two different wall panels.
[0094] Such corner casing elements have the advantage that the wall cladding ends at the corners and is therefore aesthetically pleasing for the user.
[0095] An inner side face or outer side face of the corner casing element can essentially correspond to a whole or half side face of the component. This makes it easier to form the wall with aligned casing elements, as no further casing elements of different sizes need to be used.
[0096] The system can comprise an essentially box-shaped foundation element. The foundation element can have at least one recessed and/or protruding receiving region on an upper side in an installation position, preferably two, four, six or eight recessed and/or protruding receiving regions for connection to a base element or component. On the four side faces, the foundation element can have male and/or female connecting elements which are shaped to complement each other.
[0097] The foundation element can form a rectangular or square surface. The surface can have dimensions that correspond to the lower side of the component. This has the advantage that the foundation element can be used modularly with the component.
[0098] The foundation element can have only male or only female connecting elements on one side face. This makes it easier to align the connecting elements for connection.
[0099] The foundation element can have two or more male and/or female connecting elements per side face.
[0100] The foundation element can have only male connecting elements on one side face and only female connecting elements on an opposite side face. Such an arrangement has the advantage that a user can more easily fix the foundation elements relative to each other.
[0101] The male and female connecting elements can be arranged in one installation position only in a head area of the four side faces. This has the advantage that the foundation elements only need to be raised slightly in order to connect the corresponding male and female connecting elements to each other by lowering them vertically.
[0102] The foundation element can have a cavity. The cavity can be connected to the upper side of the foundation element by a passage. This has the advantage that the cavity is accessible from above when a large number of foundation elements are connected to form a surface.
[0103] The passage may also be shaped to receive an elongate reinforcement, in particular a steel

strut or tube or, for example, a plastic strut or tube. This allows the reinforcement to be anchored in the foundation while improving the structural integrity of the system, which can form a wall, for example.
[0104] The foundation element can be open on a lower side in an installation position, or enclose the cavity on the lower side. If the foundation element is open at the bottom, a surface on which the foundation element is placed can form a lower side of the cavity. Such a cavity can save foundation element material.
[0105] A box-shaped foundation element of the system can alternatively have at least one recessed and/or protruding receiving region on an upper side in an installation position, preferably two, four, six or eight recessed and/or protruding receiving regions, for connection to a base element or the component. The four side faces of the foundation element can each have at least one undercut.
[0106] The undercut of the foundation element can extend from the upper side of the foundation element to a lower side of the foundation element. In particular, the undercut extends perpendicular to the foundation element in the installation position. The undercut can have an essentially constant cross-sectional extent over the entire extent of the undercut.
[0107] The foundation element can be closed on at least one side adjacent to the lower side. This means that the foundation formed from a large number of foundation elements can be closed to the outside so that it can be filled with filling material.
[0108] The cavity can have at least one lateral opening on one side adjacent to the lower side, so that a connection to cavities of neighboring foundation elements can be formed. Such a connection has the advantage that filling material such as concrete can be filled in one cavity and also distributed in the other cavities of other foundation elements.
[0109] The system can include a closure element for closing at least one side opening of the foundation element.
[0110] Preferably, the foundation element can have lateral openings on two, three or four sides adjacent to the lower side in the installation position to the cavity of the foundation element.
[0111] Alternatively or additionally, the system can comprise a foundation connecting element for connecting two foundation elements by engaging in the undercuts.
[0112] The at least one lateral opening of the foundation element can be at least partially formed by the undercut.
[0113] Perpendicular to the surface of the foundation element in the installation position of the foundation element, the closing element can be inserted into the lateral opening, in particular into the undercut. The closure element can be shaped complementary to the undercut so that the closure element can close a lateral opening when the closure element is inserted vertically into the undercut in the installation position of the foundation element. A length of the closing element preferably corresponds to at least one extension of the undercut or at least the lateral opening if this is formed by only a part of the undercut.
[0114] The closure element makes it possible to adjust the size of a foundation that is laid using a large number of foundation elements. In addition, a user does not have to use foundation elements with different numbers of side openings to enclose a foundation area for filling, but all foundation elements can have more than three, in particular four, side openings.
[0115] The system comprising the foundation element may comprise two foundation elements. The foundation connecting element can be shaped for essentially form-fitting attachment in two undercuts of two foundation elements in order to connect them to one another. The foundation connecting element preferably does not extend along the entire height of the foundation connecting element in the installation position. This means that a lateral opening formed by only part of the undercut can remain open even though the foundation connecting element has been inserted. The foundation connecting element can be shaped to correspond to two adjacent undercuts of two foundation elements.
[0116] In the installation position, a passage of the foundation element can extend from the upper

side through the receiving regions to the lower side of the foundation element.

[0117] The system can comprise an intermediate floor element. In an installation position, the intermediate base element has at least one attachment area on each of four vertical side faces for attachment to the receiving structure of a component. The attachment area can also be designed for attachment with an adjacent attachment area.

[0118] The intermediate floor element can have a cavity for pipes, in particular underfloor heating. Alternatively, a separate intermediate floor support element can have such a cavity.

[0119] The intermediate floor element and/or the intermediate floor support element can run continuously from one side of the wall formed by components to an opposite side of the wall.

[0120] The intermediate shelf elements and intermediate shelf support elements can be arranged in a horizontal plane at an angle relative to each other, in particular at a right angle. This arrangement increases the stability of the false floor.

[0121] The intermediate floor element and/or intermediate floor support element can rest on the casing elements and/or wall panels of the system, which are attached to the components of a wall. Thus, the intermediate floor element and/or the intermediate floor support element can be supported by the casing elements and/or wall panels.

[0122] The system can comprise a base element. In an installation position, the base element can have at least one connecting element on a lower side, which is shaped complementary to the at least one protruding or recessed area of the foundation element.

[0123] On the upper side of the base element in an installation position, the base element can have at least one recess and a substantially horizontal surface. The recess may be suitable for receiving a component or a floor covering element.

[0124] Alternatively, the base element can have at least one protruding area on the upper side.

[0125] The protruding or recessed area can be arranged vertically above the connecting element of the base element in one installation position. The recessed area or the protruding area can be shaped complementary to a male connecting element or female connecting element of the component.

[0126] The base element can comprise two, four, six or eight protruding or recessed areas. The protruding and/or recessed areas can be arranged mirror-symmetrically on one axis, preferably two axes, on the base element. The upper side of the base element can accommodate at least one floor covering element, which preferably has the same periphery as part or all of the upper side of the base element. The floor covering elements can also have multiples of the extent of the periphery of the base element, so that the amount of work required to attach them is reduced.

[0127] A vertically extending side face of the base element in the installation position can have an upwardly extending and protruding area. There may be two protruding areas on opposite sides of the base element. This protruding area can be fixed by a wall panel or a gable element.

[0128] The base element can have a passage extending from the upper side of the base element to the lower side of the base element. The passage can extend through the upwardly protruding or recessed area on the upper side and, in particular, through the connecting element located on the lower side.

[0129] The passages of the component, the foundation element and, in particular, the base element of the system can be arranged directly above one another in the respective installation position to form a continuous cavity. The elongated reinforcement can be arranged in the passage of the component, the foundation element and/or the base element, in particular in the entire continuous cavity.

[0130] The system can also include the elongated reinforcement. The elongate reinforcement may be insertable into at least two components stacked on top of each other in the passages in the installation position. The elongated reinforcement can have a length of at least 1 m, in particular at least 1.5 m, preferably at least 2 m.

[0131] This allows the reinforcement to stabilize the wall and preferably to be anchored in the

ground and/or the foundation that can be poured with filler.
[0132] The reinforcement can have a channel along a longitudinal extension of the reinforcement, which preferably extends over the entire length of the reinforcement and preferably has two openings at two longitudinal ends of the reinforcement. On the one hand, this makes it possible to save material and, on the other hand, water pipes, electrical cables or radiators can be installed in the reinforcement.
[0133] An alternative embodiment of the component can have two receiving structures as described above on the two opposite vertical side faces in one installation position. On the other two opposite vertical side faces of the component, on the other hand, the component has connection areas which are formed analogously to the connection areas of the wall panel.
[0134] Otherwise, such a component is shaped as described above. Such a component allows the casing elements to be attached directly to the component without a wall panel.
[0135] At least one side face of the component in an installation position can have a projection extending downwards or upwards.
[0136] Such an attachment allows components mounted on top of each other to be more strongly connected to each other. The attachment can extend continuously or partially between two opposing side faces of the component.
[0137] The components and components of the systems described above can each have a base area that is an integer subset of 120×80 cm.sup.2 or an integer multiple of 120×80 cm.sup.2. This has the advantage that the components can be easily transported on pallets of this size.

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## Description

[0138] The invention will now be described with reference to certain embodiments and the accompanying figures, which show:
[0139] FIGS. **1**A and **1**B: Perspective view from below and above of an embodiment of a component,
[0140] FIGS. **2**A to **2**C: Perspective view from above of three alternative embodiments of the component, FIG. **2**D: Perspective view from above of a fourth embodiment of the component connected to an attachment element,
[0141] FIGS. **3**A to **3**H: Cross-sections of various embodiments of recesses in a component,
[0142] FIGS. **4**A and **4**B: Perspective view from below and above of a first version of an attachment element,
[0143] FIGS. **4**C and **4**D: Perspective view from below and from above of a second version of an attachment element,
[0144] FIG. **5**A: Perspective view of a version of the component with a step,
[0145] FIG. **5**B: Perspective view of a version of the component according to FIG. **5**A with an attached attachment element in longitudinal section,
[0146] FIGS. **6**A and **6**B: Perspective view from below and from above of an embodiment of a foundation element,
[0147] FIGS. **6**C and **6**D: Perspective view from below and from above of an embodiment of the foundation element,
[0148] FIG. **6**E: Perspective view of an embodiment of a closure element for closing a lateral opening of the foundation element according to FIGS. **6**C and **6**D,
[0149] FIG. **6**F: Perspective view of an embodiment of a foundation connecting element for the form-fit connection of two foundation elements according to FIGS. **6**C and **6**D,
[0150] FIGS. **7**A and **7**B: Perspective view of a first and a second embodiment of the wall panel connected to a first and a second embodiment of a casing element,
[0151] FIG. **7**C: Perspective top view of a component with a third embodiment of a wall panel

attached so that a recess in the center remains free,
[0152] FIG. **8**: Perspective view of a first embodiment of a corner casing element,
[0153] FIGS. **9**A and **9**B: Perspective view of a first and second embodiment of a base element for the connection between a component and a foundation element,
[0154] FIGS. **10**A and **10**B: Perspective view of a third and fourth embodiment of a base element for the connection between a floor covering element/component and a foundation element,
[0155] FIG. **11**: Schematic, perspective view of a system with box-shaped components arranged one above the other,
[0156] FIGS. **12**A and **12**B: Perspective view of a first and second embodiment of a system comprising a component, a foundation element, a base element, a connecting element, a wall panel, and a casing element,
[0157] FIGS. **13**A to **13**C: Perspective views of a second and third embodiment of a corner casing element,
[0158] FIGS. **14**A and **14**B: Perspective views of two further alternative embodiments of a component,
[0159] FIG. **15**: Perspective view of a wall formed by a plurality of components, connecting elements for connecting the components, base elements, foundation elements with locking elements and reinforcements.
[0160] FIGS. **1**A and **1**B show perspective views of an embodiment of a component **101** from below and from above. The component **101** has two female connecting elements **7** on the lower side **5** and two male connecting elements **6** on the upper side **4**, which are arranged one above the other in an installation position **11** essentially on a vertical axis V. In addition, the male and female connecting elements **6**, **7** are arranged centrally between two opposite side faces **22**, **24**. The component **101** in the installation position **11** is twice as wide as it is deep and the connecting elements **6**, **7** are arranged exactly in the center of one half of the component **101** with a square base. The width B of the component **101** is 40 cm, the height is 20 cm and the depth D is also 20 cm. The height of the component **101** in the installation position **11**, including the male connecting element **6**, is 30 cm.
[0161] A passage **8** extends through the upper male connecting element **6** in a vertical direction and merges into the opening of the female connecting element **7** on the lower side **5**. The female connecting element **7** and male connecting element **6** are box-shaped, so that the male connecting element **6** of the component **101** can be inserted into the female connecting element **7** of a component **101** located above it in a form-fitting and rotationally secure manner. One receiving structure **3** is arranged on each of the two lateral side faces **21**, **23** and two receiving structures **3** are arranged on each of the end-face oriented side faces **22**, **24**. A receiving structure **3** comprises two recesses **31**, **32** between which a recess **37** is arranged. The longitudinal axes of the recesses **31**, **32** and the recess **37** run parallel to each other in the vertical direction of the component **101** in the installation position **11** and extend over the entire side faces **21**, **22**, **23**, **24**. The cross-section of the recesses **31**, **32** is the same along the entire vertical axis and has an undercut (see FIGS. **3**A to **3**H). The recess **37**, on the other hand, has a substantially constant rectangular cross-sectional shape along its entire longitudinal axis.
[0162] The upper side **4** of the component **101** has a step **41** extending circumferentially along the edge between the upper side **4** and the side faces **21**, **22**, **23**, **24**. The step **41** forms a partially enclosed box-shaped cavity when two components **101** are arranged on top of each other. The step **41** connects the recesses **31**, **32** and recesses **37** of the receiving structure **3** with each other. Thus, the step **41** can be used for laying, in particular in a horizontal direction in the installation position **11**, electrical or water pipes.
[0163] FIGS. **2**A to **2**C show three alternative embodiments of a cube-shaped component **102** in a perspective view. A cube-shaped form of the components **102** allows, in combination with components of double width (see FIG. **1**A and FIG. **1**B), the components to be arranged one above

the other in rows offset relative to each other (see FIG. **11**). The component in FIG. **2**A has a hollow cylindrical male connecting element **61**, so that several components **102** can be arranged one above the other in a rotatably connected manner. In contrast, the components **102** in

[0164] FIG. **2**B and FIG. **2**C have a prism-shaped male connecting element **62**. In FIG. **2**B, the male connecting element **62** has a hexagonal base and in FIG. **2**C a square base. The components **102** according to FIG. **2**B and FIG. **2**C can thus be connected in a non-rotatable manner on the upper side **4** and lower side **5** to other components **102** in an installation position **101**. The longitudinal axis of a passage **8** is aligned exactly centrally in the component **102** and runs in the vertical direction V in the installation position **11**. The same receiving structure **3** is arranged on all side faces **21**, **22**, **23**, **24**, so that no errors in the alignment of the components **102** can occur when the components **102** are arranged. Each receiving structure **3** in FIG. **2**A and FIG. **2**B comprises two recesses **31**, **32** and a depression **37**. In contrast, the receiving structure **3** of the component **102** of FIG. **2**C has only one recess **31**, which has a trapezoidal base, so that a dovetail-shaped attachment element can be attached (see FIGS. **4**C and **4**D). In addition, the recess **37** in FIG. **2**C runs from the upper side **4** to the lower side **5** in the area of two adjacent side faces **21**, **22**, **23**, **24** and has a cross-sectional area of 2 cm×2 cm.

[0165] The width B of the component **102** in FIGS. **2**A to **2**C is 20 cm, the height is 20 cm and the depth D is also 20 cm. A height of the component **102** in the installation position **11** including the male connecting element **61**, **62** is 30 cm.

[0166] FIG. **2**D shows a perspective view of a fourth embodiment of the component **101** connected to an attachment element **9** of FIGS. **4**C and **4**D in a receiving structure **3** of the component **101**. For components of the component **101** described above, reference is made to FIG. **2**C. The component **101** in FIG. **2**D has twice the width of the component in FIGS. **2**A to **2**C, 40 cm. On the two long sides **22**, **24** of the component **101** two receiving structures **3** are attached next to each other and on the shorter sides **21**, **23** one receiving structure **3** each. In the receiving structure **3** of the right short side **23** an attachment element **9** is attached. The attachment element **9** is essentially T-shaped and has a connecting bar **311** with two lateral, dovetail-shaped widenings **3111** (see FIGS. **4**C and **4**D). By inserting the connecting bar **311** into the two recesses **31** with a trapezoidal base of adjacent components **101**, the components **101** can be connected to each other. Half of a head region **93** of the attachment element **9** is arranged in a respective step **41** of the components. However, the shape of the head region **93** of the attachment element **9** leaves the step **41** of the adjacent side faces **22**, **24** free. Thus, for example, a cable can be routed horizontally to the side faces **22**, **24** within the step.

[0167] FIGS. **3**A to **3**H show cross-sections of various embodiments of recesses **32** of a component, such as according to FIG. **1**A to FIG. **2**B. All recesses **32** have an undercut **33** with a larger extent **35** than the extent **36** in an opening area **33**. The cross-section can be L-shaped as in FIG. **3**A or T-shaped as in FIG. **3**B. In addition, the cross-section can include rounded areas as in FIG. **3**C to FIG. **3**F, so that the pressure of inserted attachment elements, in particular connecting latches, can be distributed more evenly and no load peaks of the material occur. The recess **32** in FIG. **3**G and FIG. **3**H is dovetail-shaped. In all embodiments FIG. **3**A to FIG. **3**H, an attachment element can be inserted in a substantially form-fitting manner in the longitudinal direction of the recess **32**. For this purpose, two recesses **32** of two adjacent components are arranged in such a way that the side faces **22** abut against each other and the recesses **32** form a connected cavity. Subsequently, the components can be fixed relative to each other by inserting the attachment element into the cavity formed in the recesses **32**.

[0168] FIGS. **4**A and **4**B show perspective views of a first embodiment of an attachment element **9** from below and from above. In an installation position **95**, the attachment element **9** has a head region **93** which comprises lateral flanges **941**, **942** which have a wider extension **96** than a central web **92** of the head region **93**. The head region **93** of the attachment element **9** thus forms a double-T profile **91** in a horizontal plane in an installation position **95**. Such a double-T profile **91** allows the attachment element **9** to be inserted into a step on the upper side of a component in a substantially form-fitting manner (see FIG. **5**A and FIG. **5**B), so that the step prevents the attachment element **9** from tilting relative to the component.

[0169] In the installation position **95** in the laterally outer region of the flanges **941**, **942**, the head region **93** is materially connected to a first connecting latch **311** and a second connecting latch **321**. A center bar **371** for connecting the recesses is arranged between the first connecting bar **311** and the second connecting bar **321**. The connecting bars **311**, **321**, **371** run parallel and are not connected to one another on the side opposite the head region **93**. Thus, the connecting latches **311**, **321**, **371** can be inserted together from above into the recesses of two adjacent components in the installation position **95** by a single translational movement. FIG. **4**A also shows that the first and second connecting latches **311**, **321** are c-shaped in cross-section and the open sides **312**, **322** of the c-shaped connecting latches **311**, **321** are aligned with each other. Such an inwardly directed shape **312**, **322** of the recesses **311**, **321** allows the recesses **311**, **321** to be placed laterally far apart, so that the fixation with an attachment element **9** can be improved. Two C-shaped connecting bars **311**, **321** make the connection of the components more flexible. Even if the connecting latches **311**, **321** are damaged/defective, a connection can still be ensured by the other connecting latch **311**, **321** of two components.

[0170] FIGS. **4**C and **4**D show perspective views of a second embodiment of an attachment element **9** from below and from above. In an installation position **95**, the attachment element **9** has a head region **93** which has a constant extension **96**. The attachment element **9** also has a central connecting bar **311**. The connecting bar **311** has two opposing dovetail-shaped widenings **3111** on each of the lateral side faces **98**, **99**. The connecting latch **311** of the attachment element **9** can fix the two components relative to each other by a translational movement into the recesses of two adjacent components with trapezoidal recesses (FIG. **2**C or FIG. **2**D). The connecting bar **311** has a substantially congruent shape to the trapezoidal recesses with undercut of the components. Such a dovetail-shaped connecting bar **311** makes the connection of two components more stable and rigid.

[0171] FIG. **5**A shows a perspective view of an embodiment of a component **102** with a step **41** highlighted by a dotted area. In an installation position **11** of the component, the step **41** extends from an upper end of a first recess **31** via the upper end of a recess **37** to the upper end of a second recess **32** of the receiving structure **3**. All side faces **21**, **22**, **23**, **24** have identical receiving structures **3** with recesses **31**, **32** and the recess **37**, at the upper end of which the step **41** of the upper side **4** is arranged in an installation position **11**.

[0172] FIG. **5**B shows a perspective view of an embodiment of the component **102** according to FIG. **5**A with an attachment element **9** inserted into the receiving structure **3** in longitudinal section LS (see FIG. **4**B). The longitudinal section LS runs exactly centrally through the attachment element **9** parallel to the side face **22**. One half of a head region **93** of the attachment element **9** with a double-T profile **91** is marked with dots in FIG. **5**B for better visibility. The head portion **93** of the attachment element **9** positively fills the step between a side **22** and the upper side **4** of the component (see FIG. **5**A). The attachment element **9** also has connecting latches **311**, **321**, **371**, which are also positively fixed in the recesses **321**, **322** and a recess **37**. The recesses **32**, **32** and the recess **37** are arranged adjacent to the recesses **32**, **32** and the recess **37** of the connected component **102** in the connected state of two components **102**. The height H, width B, and depth D have the same dimensions as in FIGS. **2**A and **2**B.

[0173] FIGS. **6**A and **6**B show a foundation element **17** in a perspective view from below and from above. The foundation element **17** has a cavity **179** on a lower side **1712**, which is suitable for receiving fillers such as cement or reinforcement such as steel struts. In addition, the cavity **179** of the foundation element **17** is suitable for accommodating pipes, in particular water pipes of an underfloor heating system. The cavity **179** has an opening **1791** to all side faces **173**, **174**, **175**, **176** of the foundation element **17**. The openings **1791** are arranged centrally on the side faces **173**, **174**,

**175**, **176** so that the cavities **179** of adjacent foundation elements **17** can be connected to one another. On the upper side **1711**, the foundation element **17** also has a passage **1792**, which is connected to the cavity **179**. Thus, the foundation element **17** can be filled from above, for example with cement through the passage **1792** after proper attachment.

[0174] In an alternative embodiment, the cavity **179** can also be closed at the bottom in an installation position **171** of the foundation element. In addition, the foundation element **17** has female connecting elements **178** or male connecting elements **177** on the edges of the upper side **1711** and the side faces **173**, **174**, **175**, **176**.

[0175] In FIG. **6**A and FIG. **6**B, a pair of male or female connecting elements **177**, **178** are each arranged on a side face **173**, **174**, **175**, **176**. The male and female connecting elements **177**, **178** extend from the upper side only over a partial area of the side faces **173**, **174**, **175**, **176** and not completely to the lower side **1712**. Thus, adjacent foundation elements **17** in the installation position **171** can be connected to each other by the complementary female and male connecting elements **177**, **178** even by slightly lifting one of the foundation elements **17**. The upper side **1711** also has four recessed receiving regions **172** so that a base element can be attached for connection to a floor covering element or component (see FIGS. **9** and **10**). The recessed receiving regions **172** are arranged on the foundation element **172** in such a way that the receiving regions **172** of several connected foundation elements **17** are equidistant from one another. Thus, for example, components (see FIGS. **1**A and **1**B) can be attached to two receiving regions on one foundation element **17** or to two adjacent foundation elements **17**. The foundation element **17** has a height of 30 cm, a depth of 40 cm and a width of 40 cm. Such a size provides an ideal weight for attaching the foundation elements **17**, as they are easy to lift and still cover a large area. The base area in FIGS. **6**A and **6**B is thus exactly twice as large as the base area of a component in FIGS. **1**A, **1**B or **2**D. Due to the large cavity **179**, the foundation elements **17** are nevertheless easy to lift in order to be placed.

[0176] FIGS. **6**C and **6**D show a perspective view from below and from above of an embodiment of the foundation element **170**. The foundation element **170** has four side faces **1703**, **1704**, **1705**, **1706**, each of which has two undercuts **1707**. The undercuts **1707** extend vertically in an installation position **1701** (see FIG. **6**D) from an upper side **17011** to a lower side **17012** of the foundation element **170**. The foundation element **170** also has four receiving regions **1702** on the upper side **17011**, which can be connected to a base element **18** (see FIGS. **9**A-**10**B). The receiving regions **1702** can also alternatively form passages for receiving reinforcements **20** (see FIG. **15**).

[0177] FIG. **6**C shows that the foundation element **170** forms an open cavity **1709** on the lower side **17012**, each of which is connected to lateral openings that are partially formed by the undercuts **1707**. Thus, the cavity **1709** can be connected to adjacent cavities of other foundation elements **170** so that, for example, a foundation of fillers such as cement can be poured. The foundation element **170** also has a passage **1710** on the upper side **17011**, so that the foundation element **170** can be filled from above in the installation position **1701**.

[0178] To prevent the fillers from escaping, for example at the edge of the planned foundation, a closing element **1708** can close the lateral opening in the form of the undercut (see FIG. **6**C). Neighboring foundation connecting elements **170** can also be connected with a foundation connecting element **90**, which can be positively inserted into the undercuts **1707** of neighboring foundation connecting elements **170** (see FIG. **6**F).

[0179] FIG. **6**E shows a perspective view of an embodiment of a closure element **1708** for closing a lateral opening of an embodiment of the foundation element **170** according to FIGS. **6**C and **6**D. A cross-sectional area of the closure element **1708** is essentially isosceles trapezoidal and is constant over an entire longitudinal extent L of the closure element **1708**. This enables the closure element **1708** to be easily inserted into the undercut **1707** (see FIGS. **6**C and **6**D) in order to close the cavity **1709** located behind it, so that no filler can escape during filling.

[0180] FIG. **6**F shows a perspective view of an embodiment of a foundation connecting element **90** for positively connecting two foundation elements **170** according to FIGS. **6**C and **6**D. A cross-sectional area of the foundation connecting element **90** is constant along a longitudinal extension L of the foundation connecting element **90**. The cross-sectional area is formed by two isosceles trapezoids, which are connected to each other at their short side (see dashed line). In addition, the foundation connecting element **90** has a projection **100** in an upper region in an installation position, which is formed to correspond to a recess in the foundation connecting element **170** (see FIGS. **6**C and **6**D).

[0181] FIGS. **7**A and **7**B show two perspective views of a first and a second embodiment of a wall panel **12**, which is connected to a casing element **15**. The wall panel **12** in FIG. **7**A has a plurality of first L-shaped connection areas **131** on a first side face **13**. The open sides of the L-shaped connection areas **131** face each other in pairs, so that an intermediate space **121** is partially enclosed. On each of the first side faces **13**, three such L-shaped connection areas **131** are arranged in pairs in two adjacent rows **133**, **134**.

[0182] Second L-shaped connection areas **141** are arranged on an opposite second side face **14** of the wall panel **12** in FIG. **7**A and FIG. **7**B. The second L-shaped connection areas **141** extend over a large part (see FIG. **7**B) or the entire second side face **14** (see FIG. **7**A) and run orthogonally to the first connection areas **131**. In an installation position of the wall panel **12**, the first connection areas **131** are oriented with their longitudinal axis vertical and the second connection areas **141** are oriented with their longitudinal axis horizontal. The wall panel **12** also has a plurality of stepped regions **122**, so that adjacent wall panels **12** contact each other in a larger area and thus seal and insulate better. A separate seal on the step-shaped areas **122** would also be conceivable.

[0183] The first L-shaped connection areas **131** can be positively inserted into the recesses **31**, **32** with an undercut of the receiving structure **3** of the component (see FIGS. **1**A and **1**B) in order to be connected to a side face of the component. The recesses of components arranged one above the other are connected to each other at the sides, so that the wall panel **12** with the first connection area **131** can be inserted along the recesses of several components in an installation position from above.

[0184] A first row **133** of the connection areas **131** of the wall panel **12** can thereby be connected to a first component, and a second row **134** of the connection areas **131** of the wall panel **12** can be connected to a second component. Thus, the amount of work required to attach the wall panels can be reduced as a larger area can be attached in a single step. However, the central recess (see e.g. FIGS. **1**A and **1**B, **37**) is left free when the wall panel according to FIG. **7**A is attached, so that a cable can be routed therein (see FIG. **8**).

[0185] In a second embodiment of a wall panel **12** in FIG. **7**B, the first connection areas **132** are dovetail-shaped on an upper side **13**. The dovetail-shaped connection areas **132** can thus be inserted into an undercut of the recess (see FIGS. **2**C and **2**D, **31**) with a trapezoidal base surface in a substantially form-fitting manner.

[0186] On the upper side **13**, two connection areas **132** are arranged next to each other in two rows **133**, **134**.

[0187] The second L-shaped connection areas **141** in FIGS. **7**A and **7**B are shaped to positively receive an L-shaped receiving region **151** of the casing element **15**. For this purpose, the receiving region **151** is shaped complementary to the second connection area **141**. In an installation position of the wall panel **12**, an opening of the L-shaped second connection area **141** preferably points upwards, so that the receiving regions **151** of the casing element **15** can be inserted from above.

[0188] An outer side **16** of the casing element **15** is substantially aligned, so that adjacent casing elements **15** form a substantially continuous and aligned surface.

[0189] FIG. **7**C shows a perspective top view of a third embodiment of a wall panel **12** attached to a component **101** according to FIGS. **1**A and **1**B. The wall panel **12** in FIG. **7**C has an aligned first side face **13** and a second side face **14** with two L-shaped connection areas **141**. The two L-shaped connecting elements **141** are substantially positively inserted into recesses **31**, **32** of the component

**101** from an upper side **4** of the component **101** so that the wall panel **12** is fixed to the component **101**. In addition, the wall panel **12** has a stepped area **122** for increasing the contact surface to adjacent wall panels **12**. However, a recess **37** of the size 2 cm×2 cm of the component **101** remains free despite the wall panel **12** being attached. Thus, an electricity or water pipe can run or be attached in the recess **37**. The components **101** are shaped in such a way that the recesses **37** on a side face **24** of components **101** arranged one above the other connect to each other in an installation position in order to guide vertically running lines. In addition, the L-shaped connection areas in FIG. **7**C do not extend over the entire side face **14**, so that a step **41** of the component **101** remains free. This allows the cables to also be routed horizontally along the step **41** in the installation position of the component.

[0190] FIG. **8** shows a perspective view of a first embodiment of a corner casing element **152** with two connection areas **153** on each inwardly facing side face **157**, **158**. The connection areas **153** are each connectable to the L-shaped connection areas of the wall panels (see FIGS. **7**A, **141**) with two side faces of a component arranged next to each other and orthogonally. The connection areas **153** run horizontally so that the L-shaped opening can be aligned downwards. Thus, the corner casing element **152** can be inserted vertically from above into the connection areas of the wall panel. Such a corner casing element **152** has two flat surfaces, which are formed by two outer surfaces **155**, **156** of the corner casing element **152**. In FIG. **8**, the outer surfaces **155**, **156** form an angle of 270° to each other. The inner surfaces **157**, **158**, on the other hand, form an angle W of 90° to each other in FIG. **8**. Thus, the corner casing element **152** can be attached to an outer corner with adjacent sides at 270° to each other of a component system.

[0191] Analogous to this embodiment of the corner casing element **152**, in a second embodiment the connection areas **153** can also be arranged on the outer surfaces **155**, **156** to form a corner by two aligned inner surfaces **157**, **158** of 90 degrees (not shown in the figures). Thus, the corner casing element can be attached to an inner corner with adjacent sides at 90 degrees to each other.

[0192] FIGS. **9**A and **9**B show a perspective view of a first and second embodiment of a base element **18** for connection between a component **101** (see FIGS. **1**A and **1**B or FIGS. **2**C and **2**D) and a foundation element **17**, **170** (see FIGS. **6**A to **6**D). The base element **18** has a protruding box-shaped region **185** on an upper side **183** in an installation position **181**, which is shaped for positive connection with a female connecting element **7** (see FIG. **1**A) of the component. On a lower side **182** of the base element **18** in the installation position **181**, the base element **18** has two box-shaped male connecting elements **184**. The male connecting elements **184** are shaped complementary to the recessed areas **172**, **1702** (see FIG. **6**B or FIG. **6**D) of a foundation element **17**, **170** in FIGS. **6**A-**6**D. In addition, the base element **18** in FIGS. **9**A and **9**B has dovetail-shaped connecting elements **1811**, **1812** on all four side faces **186**, **187**, **188**, **189**.

[0193] FIG. **9**A shows that a female dovetail-shaped connecting element **1812** is arranged on each of the two front side faces **186**, **187** and a male dovetail-shaped connecting element **1811** is arranged on each of the rear side faces **188**, **189**. Thus, horizontally adjacent base elements **18** can be connected to each other by the dovetail-shaped connecting elements **1811**, **1812** in FIG. **9**A.

[0194] FIG. **9**B shows that a female dovetail-shaped connecting element **1812** is disposed on each of the side faces **186**, **187**, **188**, **189**. The female connecting elements **1812** can be connected to each other by base connecting elements, which are shaped similarly to the foundation connecting elements **90** (see FIG. **6**F).

[0195] In addition, the base element **18** in FIGS. **9**A and **9**B has a passage **1813** which, when properly installed, is adjacent to the passage (see, for example, FIGS. **1**B, **8**) of the components.

[0196] FIGS. **10**A and **10**B show a perspective view of a third and fourth embodiment of a base element **19** for connecting a floor covering element, in particular parquet flooring, and a foundation element (see FIGS. **6**A and **6**B). In contrast to FIGS. **9**A and **9**B, the base element **19** has a connecting element **194** on an upper side **193** in the form of a recessed box-shaped region **194**. The recessed area **195** is shaped in a complementary manner for receiving a floor covering element (not

shown in the figures).
[0197] The recessed region **195** may also be shaped complementary to receive a male connecting element **6** of the component (see FIGS. **1**A and **1**B).
[0198] The dovetail-shaped connecting elements **1911**, **1912** in FIG. **10**A on the side faces **196**, **197**, **198**, **199** of the base element **19** and the male connecting element **194** are shaped analogously to FIG. **9**A.
[0199] The dovetail-shaped connecting elements **1912** in FIG. **10**B on the side faces **196**, **197**, **198**, **199** of the base element **19** and the male connecting element **194** are shaped analogously to FIG. **9**B.
[0200] A base area of the base element **18**, **19** in FIG. **9**A to FIG. **10**B is 40 cm×20 cm, so that two base elements **18**, **19** can be arranged on the foundation element to cover it completely.
[0201] FIG. **11** shows a schematic, perspective view of a system **1** with several components **101**, **102** that are partially offset from each other in order to increase stability. The stability can also be further increased, for example by passing steel struts through adjacent passages **8** (see FIGS. **1**A and **1**B) through several components **101**, **102** arranged one above the other. For better clarity, the receiving structures **3** and connecting elements **6**, **7** (see FIGS. **1**A and **1**B) of the components **101**, **102** are not shown in FIG. **11**.
[0202] The system **1** in FIG. **11** comprises both essentially box-shaped components **102** with a base area of 20 cm×20 cm and box-shaped components **102** with a base area of 20 cm×40 cm. The height of the components **101**, **102** is 30 cm.
[0203] FIGS. **12**A and **12**B show a perspective view of a first and second embodiment of a system **1** comprising a construction element **101**, a foundation element **17**, a base element **18**, a connecting element **9**, a wall panel **12**, and a casing element **15**. On a half base surface of the foundation element **17**, the base element **18** is fixed by two male connecting elements **184** (see FIG. **10**) on the lower side in recessed and box-shaped areas **172** of the foundation element **17**. On the upper side of the base element **18**, the base element **18** is connected to the female connecting elements **7** (see FIG. **1**A) by inserted box-shaped areas **185** (see FIG. **10**). The base element **18** and the component **101** have the same base area of 40 cm×20 cm.
[0204] On a front side **21** of the component **101** in FIGS. **12**A and **12**B, an attachment element **9** is half recessed in a receiving structure **3** of the component **101**, so that the head portion **93** (see FIGS. **4**A-**4**C) fills a step **41** (see FIG. **1**B) of the component **101** on the front side **21**. In FIG. **12**A, the component **101** has a double L-shaped receiving structure according to FIGS. **2**A and **2**B and in
[0205] FIG. **12**B a partially dovetail-shaped receiving structure according to FIGS. **2**C and **2**D. In FIG. **12**A, the attachment element **9** has two lateral connecting latches **311**, **321** and a central latch **371** (see FIGS. **4**A and **4**B), analogous to FIGS. **4**A and **4**B. In FIG. **12**B, on the other hand, the attachment element **9** has a dovetail-shaped central connecting bar **311** (see FIGS. **4**C and **4**D). The wall panel in FIG. **12**A has two double L-shaped first connection areas **131**, shaped complementary to the receiving structure **3** of the component **101**, analogous to FIG. **7**A. In FIG. **12**B, on the other hand, the first connection areas **131** are dovetail-shaped and complementary to the receiving structure **3**, analogous to FIG. **7**B.
[0206] In FIGS. **12**A and **12**B, the casing elements **15** are inserted into the upwardly open second L-shaped connection areas **141** to secure the casing elements **15** to the wall panel **12**. In FIGS. **12**A and **12**B, the wall panels **12** are twice as high as the component **101** without a male connecting element **6** of the component **101**. Such wall panels **12** achieve an optimum weight of the wall panels **12** and also reduce the labor required for attaching them to the components **101**. Since the receiving structure **3** of the components **101** extends vertically upwards in the installation position, several receiving structures **3** can be connected to each other so that the wall panel **12** can be easily attached.
[0207] FIGS. **13**A to **13** C show a second and a third embodiment of a corner casing element **152** with two adjacent connection areas **153** at an angle to each other. In the second embodiment of the

corner casing element **152** in FIG. **13**A, the connection areas **153** are arranged in four rows one above the other on the inwardly facing side faces **157**, **158**. In the third embodiment of the corner casing element **152** in FIG. **13**B and FIG. **13**C, the connection areas **153** are arranged in only two rows one above the other on the inwardly facing side faces **157**, **158**. The connection areas **153** can each be connected to the connection areas **141** of the wall panels (see FIG. **7**A) or directly to a component (see FIG. **14**). Analogous to FIG. **8**, the connection areas **153** have L-shaped openings which can be aligned vertically downwards in an operating position in order to be connected to the wall panels and/or components. The adjacent connection areas **153** of the corner casing element **152** are arranged at an angle W of 90° to each other. However, an analogous embodiment of 270° would also be conceivable for shuttering internal corners. By such an embodiment, the corner casing element **152** can be attached to an inner corner of a wall with adjacent sides at 90° to each other.

[0208] The corner casing element **152** in FIG. **13**A has four connection areas **153** arranged one above the other and the corner casing element **152** in FIG. **13**B and FIG. **13**C has two connection areas **153** arranged one above the other, so that the corner casing element can be connected to a plurality of wall panels or components at the same time. For this purpose, the four connection areas **153** are rigidly connected to each other by an elongated support **159**. In this way, the amount of work required to attach the corner casing element **153** can be reduced and, in addition, a casing that is as flat as possible can be achieved.

[0209] For better visibility of the connection areas **153**, an exemplary structured shuttering **160** has been shown in FIG. **13**A on only one outer side **155**. However, this casing **160** is attached to the corner casing element **152** on both outer sides **155** and **156** and is rigidly connected thereto.

[0210] FIGS. **14**A and **14**B show perspective views of two further alternative embodiments of a component **101** in an installation position **11**. The component **101** differs from the component in FIGS. **1**A and **1**B by dovetail-shaped receiving structures **3** with only one recess **31** on each of the two end faces **21**, **23**. In addition, the component **101** in FIG. **14**A and FIG. **14**B has upwardly open L-shaped receiving structures **38** on the two lateral side faces **22**, **24**. These L-shaped openings of the receiving structures **38** run horizontally in the installation position **11** and extend over an entire longitudinal length of the component **101**. A casing element **16** (see FIGS. **7**A and **7**B) can be attached directly in these receiving structures **38** without requiring a wall panel **12** (see FIG. **7**A). In addition, these receiving structures **38** also have a vertically extending recess **37** with a rectangular cross-sectional area for guiding cables. A further difference between the embodiment of the component **101** in FIG. **14**A and FIG. **14**B and FIGS. **1**A and **1**B is that the component **101** has a downwardly protruding projection **39** on each of the lateral side faces **22**, **24** and a receptacle **40** for this projection **39**. This projection **39** extends over the entire longitudinal length of the component **101**. The projection **39** is adapted to substantially positively enclose an underlying side face **22**, **24** of another component **101** and to be inserted into the receptacle **40**. By means of the attachment **39** and the receptacle **40**, a more stable mounting of components **101** arranged one above the other can be achieved. A renewed description of the previously described features (see FIG. **1**A, FIG. **1**B, FIG. **5**A and FIG. **5**B) has been omitted in FIGS. **14**A and **14**B.

[0211] The embodiment of the component **101** in the installation position **11** in FIG. **14**B differs from the embodiment in FIG. **14**A by a horizontal recess **42**, which is arranged on a lateral side face **22** and extends partially through the component **101**. This recess **42** is adapted to receive a steel beam as a supporting device for an intermediate floor. Thus, by such components **101** having horizontal recesses **42** on opposite wall portions, steel beams can be received so that a false floor can be supported by steel beams.

[0212] FIG. **15** shows a perspective view of a wall formed by a plurality of components **101** (see FIG. **2**D), connecting elements **9** for connecting the components **101**, base elements **18** (see FIGS. **9**A-**9**B), foundation elements **170** (see FIGS. **6**C and **6**D) with closure elements **1708** (see FIG. **6**E) and elongated reinforcements **20**.

[0213] The side faces **1703**, **1704** of the foundation element **170** are sealed by a plurality of closure elements **1708** so that the adjacent cavities **1709** can be filled with filler material through the passages **1710** (see FIGS. **6**C and **6**D).
[0214] In contrast to FIGS. **6**C and **6**D, the foundation element **170** also has passages in the receiving regions **1702** so that the reinforcement **20** can be guided through the receiving elements.
[0215] The elongated reinforcement **20** is formed by a PVC or steel pipe, which can be passed through the adjacent passages of the component **101**, base element **18** and foundation element **170**. This enables the reinforcement to be anchored in the ground and/or the foundation which can be cast using filler material.

## Claims

**1-18**. (canceled)
**19**. A box-shaped component comprising at least partly plastic and four side faces which are aligned vertically in an installation position of the box-shaped component and each have at least one receiving structure for receiving at least one attachment element, wherein an upper side of the box-shaped component in the installation position has a male connecting element and a lower side has a female connecting element, or the upper side has a female connecting element and the lower side has a male connecting element, wherein the male connecting element and female connecting element are shaped complementary to each other.
**20**. The box-shaped component according to claim 19, wherein the at least one receiving structure extends continuously in a straight line from an upper side of the component to a lower side of the component.
**21**. The box-shaped component according to claim 19, wherein the receiving structure comprises at least one recess extending over a part or the entire side face, wherein a cross-section of the recess comprises an undercut, and the cross-section of the recess is L-shaped or dovetail-shaped.
**22**. The box-shaped component according to claim 21, wherein the receiving structure has at least two recesses and the longitudinal axes of the recesses run parallel and spatially spaced from one another.
**23**. The box-shaped component according to claim 19, wherein the upper side comprises a step between at least one of the side faces and the upper side, so that box-shaped components arranged one above the other form an at least partially enclosed cavity which is adjacent to a recess.
**24**. The box-shaped component according to claim 19, wherein the box-shaped component comprises a passage having a longitudinal axis extending in a straight line from the upper side to the lower side.
**25**. The box-shaped component according to claim 19, wherein, in the installation position of the box-shaped component, the box-shaped component has a height H, measured in the vertical direction, in the range from 10 cm to 40 cm; a width W, measured from the farthest side faces, in the range from 10 cm to 80 cm; a depth D, measured in a direction perpendicular to the height and width, in the range from 10 cm to 40 cm.
**26**. A method of manufacturing a box-shaped component according to claim 19, wherein the box-shaped component is manufactured by injection moulding.
**27**. A system comprising at least two components according to claim 19 and an attachment element for positive connection to a receiving structure of one of the components and for connecting two receiving structures of the components.
**28**. The system according to claim 27, wherein the system comprises at least one first box-shaped component and at least one second box-shaped component, wherein the first box-shaped component is half as large, one third as large, or one quarter as large as the second box-shaped component.
**29**. The system according to claim 27, wherein the system comprises a wall panel, and the wall

panel has a first connection area on a first side face, which protrudes from the side face and can be positively connected to a receiving structure of the box-shaped component.

**30**. The system according to claim 29, wherein the system comprises a separate casing element, wherein the wall panel has a second connection area (**141**) for attaching the casing element on a second side face opposite the first connection area.

**31**. The system according to claim 30, wherein the system comprises a corner casing element, wherein the corner casing element has at least one connection area, so that the corner casing element is connectable to at least two receiving structures of adjacent side faces of a box-shaped component or is connectable to two second connection areas of two different wall panels.

**32**. The system according to claim 27, wherein the system comprises a box-shaped foundation element, which has on an upper side in an installation position at least one of a recessed and/or a protruding receiving region for connection to a base element, and has male connecting elements and female connecting elements on the four side faces, which are shaped complementary to one another.

**33**. The system according to claim 27, wherein the system comprises a substantially box-shaped foundation construction element which, on an upper side in an installation position, has at least one of a recessed and/or a protruding receiving region for connection to a base element or the box-shaped component, and on each of the four side faces at least one respective undercut.

**34**. The system according to claim 33, wherein the system comprises a closing element for closing at least one lateral opening of the foundation connecting element in the installation position, to a cavity of the foundation element and/or a foundation connecting element for connecting two foundation elements by engaging in the undercuts.

**35**. The system according to claim 32, wherein the system comprises a base element which, in an installation position, has on a lower side at least one connecting element which is shaped complementary to the at least one protruding region or the recessed region of the foundation element and has on the upper side has at least one recessed region and a horizontal surface or has at least one protruding region on the upper side, wherein the recessed region or the protruding region is shaped complementary to a male connecting element or a female connecting element of the box-shaped component.

**36**. A box-shaped component comprising at least partly plastic and four side faces aligned vertically in an installation position of the box-shaped component, wherein at least two side faces have a receiving structure for receiving at least one attachment element and wherein at least one further receiving structure has a recess for receiving a cable, which extends from one side face to an opposite side face.

**37**. The box-shaped component according to claim 36, wherein an upper side of the box-shaped component in the installation position has a male connecting element and a lower side has a female connecting element, or the upper side has a female connecting element and the lower side has a male connecting element, wherein the male connecting element and female connecting element are shaped complementary to each other.

**38**. The box-shaped component according to claim 25, wherein, in the installation position of the box-shaped component, the box-shaped component has a height H, measured in the vertical direction, in the range from 15 cm to 25 cm; a width W, measured from the farthest side faces, in the range from 15 cm to 50 cm; a depth D, measured in a direction perpendicular to the height and width, in the range from 15 cm to 25 cm.